Lecture 22: Naval Nuclear Operations

CBE 30235: Introduction to Nuclear Engineering
Wednesday, March 4, 2026

1. The Operational Record

The U.S. Naval Nuclear Propulsion Program (NNPP) is widely considered the most successful high-consequence safety program in history. Its safety record is built on a foundation of extreme technical discipline, known as the "Rickover" philosophy, which mandates total accountability for every component and procedure.

1.1 Core Safety Statistics

Since the program's inception in 1948, the Navy has managed a massive industrial footprint:

- **Total Reactors Operated: 273 reactor plants** historically.
- **Total Reactor Cores: 562 cores** have been taken "critical" (started up).
- **Design Diversity:** The program has developed **33 different reactor designs**, from the massive 8-reactor plant of the original *USS Enterprise* to the "life-of-ship" cores in the new *Virginia* and *Columbia* class submarines that never require refueling.
- **Total Reactor-Years:** Over **7,500 reactor-years** of safe operation.
- **Miles Steamed:** More than **177 million miles** traveled on nuclear power—equivalent to over 370 round trips to the moon.
- **Radiological Accidents: Zero.** The U.S. Navy has never experienced a reactor accident, a meltdown, or any release of radioactivity that had an adverse effect on human health or marine life.
- **Active Reactors:** The Navy currently operates **99 reactors** across **79 warships** (including 11 aircraft carriers and 68 submarines). This equates to approximately 16% of all operating reactors worldwide

1.2 Personnel & Occupational Safety

The shielding on U.S. naval vessels is so effective that the average crew member actually receives a lower annual radiation dose than a civilian living on land.

- **Comparison of Annual Exposure (Average):**
  - **U.S. Navy Sailor:** ~0.038 rem (due to heavy shielding from cosmic and terrestrial radiation).
  - **Civilian Background Radiation:** ~0.300 rem (mostly from radon and the sun).
  - **Commercial Airline Crew:** ~0.170 rem (due to higher cosmic ray exposure).
- **Occupational Health:** For over 50 years, no personnel in the program have exceeded the federal annual limit for radiation exposure.
- Radiation exposure is measured in REM (Roentgen Equivalent Man) and is the measure of biological damage caused by ionizing radiation (effects on the human tissue)

- More in Lecture 35 (April 15th)

1.3 The Training Pipeline

Every Naval Nuclear operator goes through the following training pipeline:

- 6 months of classroom training learning:
    - **Nuclear Physics** (Neutron lifecycles, cross-sections)
    - **Heat Transfer and Fluid Flow** (Thermodynamics)
    - **Reactor Dynamics** (Xenon transients, reactivity coefficients)
    - **Radiological Controls** (Health physics)
    - **Chemistry and Materials** (Corrosion control)

  To pass, you must be successful complete comprehensive written exams and oral interviews.

- 6 months of operating an actual land based reactor where you get initial certification as a nuclear operator. During the 6 months, there are multiple written exams and oral interviews. The final exam sequence consists of:
    - **Observed routine and casualty watch**
    - **Comprehensive written exam**
    - **2 hour oral interview**

- 3-6 months of shipboard operations
    - **Observed routine and casualty watches**
    - **Comprehensive written examination**
    - **Oral interview by Engineer Officer**
    - **Oral interview by Commanding Officer**

2. Philosophy of Operations

2.1. U.S. Navy: Verbatim Compliance

"Verbatim Compliance" means that if a written procedure tells you to turn a valve clockwise, you turn it clockwise. If the procedure is wrong, you do not "fix" it on the fly; you stop the evolution and get the procedure formally changed.

- **The Philosophy:** As in any complex system or organization, humans are the weakest link. Naval power plants and their associated procedures involve countless years of engineering and research. Each one is thoroughly tested prior to being authorized for use in the "at sea" environment. Operators are expected to have expert level knowledge of these procedures so that in casualty or combat environments, there is perfect execution of them – no deviations. Verbatim Compliance with these procedures ensures the reactor is always placed in a safe state – whether operating or shutdown.

- **Verbatim Compliance does not equal Blind Obedience.** Sailors are taught "Backup" and "Forceful Watchteam Supervision." If a supervisor orders something that violates a procedure, the junior sailor is expected to stop them.

2.2. Civilian: Engineering Judgment

In a commercial plant, while procedures are strictly followed, there is a much higher reliance on **Engineering Judgment**. This is the application of technical expertise, historical data, and professional experience to determine the best course of action when a situation isn't "black and white."

- **The Philosophy:** A commercial plant is a massive, aging industrial asset. Unlike a standardized class of submarines, every civilian plant is slightly different. Operators and engineers are expected to understand the "Licensing Basis"—the *why* behind the limit—and use their judgment to keep the plant within its safety envelope.
- **Flexibility in Problem Solving:** If a pump is vibrating strangely, a civilian engineer might use "judgment" to determine if it can run for another 24 hours based on trending data. In the Navy, if the procedure requires the pump secured – the pump is secured. .
- **The "Why" Factor:** If an operator understands the *intent* of a step (e.g., "this step is meant to protect the condenser"), but they see that the condenser is already isolated, they can recognize that performing the step might actually starve a different system of cooling water.

2.3 Why the difference?

The different approaches stem from approaches to risk management and specific operating environment. The Navy prioritizes **predictability and combat readiness**, while civilian plants prioritize **economic longevity, continuity of power, and site-specific technical depth**

- **The Navy (Verbatim):** Standardized designs allow for standardized procedures. No design or procedure makes it to the fleet without rigorous evaluation and testing. Operators are expected to understand the basis and rationale for whatever procedure they are performing.
- **Civilian (Judgment):** Most civilian plants were built as "one-offs." Even "sister plants" have different piping layouts, different turbine manufacturers, and 40 years of different maintenance histories. Because the plants aren't identical, the local engineers **must** be the experts on their specific systems. They are able to modify actions so long as they are meeting the intent of the intended procedure.

3. Reactor Design and Operations for a Submarine

3.1 The Engineroom and Reactor Compartment Design

ADM Rickover was the driving force that determined how to fit a nuclear reactor and the associated propulsion plant inside the space of a submarine engine room. He didn't just "shrink" a power plant; he had to rethink the entire physics of how a reactor operates to move it from a sprawling land-based facility into a 28-foot-wide, 100 foot-long steel tube.

3.1.1. The Fuel: Weapons Grade Uranium-235

Commercial land reactors often use low-enriched uranium. To keep the reactor core small enough to fit in a submarine, **highly enriched Uranium-235** was selected as the fuel source.

- **The Benefit:** This allowed for a much smaller "critical mass." A small core of highly enriched fuel can produce the same amount of heat as a massive core of low-enriched fuel.
- **The Result:** A reactor core roughly the size of a **large trash can** could power a 3,000-ton vessel for years.

3.1.2. The Shielding "Sandwich"

In a land-based plant, you can use 10 feet of concrete to stop radiation. On a sub, concrete is too heavy and bulky. They developed a "shielding sandwich" to save space:

- **Water and Polyethylene:** To slow down and stop neutrons.
- **Lead and Steel:** To block gamma rays.
- By layering these materials tightly around the reactor vessel, they reduced the shielding footprint by thousands of square feet while still keeping the crew safe.

Layer	**Material**	**Primary Target**	**Why?**
Inner	Water / Steel	Fast Neutrons	Slows them down (thermalization).
Middle	Lead	Gamma Rays	Absorbs high-energy radiation from the core.
Outer	Polyethylene	Thermal Neutrons	Catches the "slow" neutrons that escaped the inner layers.

3.1.3. The PWR (Pressurized Water Reactor) Choice

Rickover chose the PWR design specifically for its density. In this system, water is kept under such immense pressure that it **cannot boil**, even at temperatures well above 100◦C (usually around 300◦C).

- Liquid water is much denser than steam, it carries heat away from the core far more efficiently.
- This allowed the heat exchangers (the "boilers") to be significantly smaller than they would be in any other type of system.

3.2 Making it last a "lifetime"

- Refueling can take up to 4 years (compared to 30 days for a commercial plant).
- Design and operations work together
  - Highly enriched U-235 (Weapons grade, >90%)
  - Reflector surrounds the core (neutron efficiency)
  - Burnable poisons loaded into the core (flattens the flux)
  - Fuel plates vs. fuel pins (more efficient heat transfer)
  - Materials that have good corrosion resistance (Zirconium, Hafnium, etc)
  - Constant Tave programming (reduces Thermal Stress at the expense of turbine efficiency)
  - Operating procedures
    - Track operating power levels (estimate fuel depletion)
    - Routinely measure core reactivity (validate fuel depletion)
    - Operational assignments are informed by "gas in the tank"

3.3 Nuclear Operations at Sea

3.3.1 Accounting for Iodine and Xenon

Due to the small core volume and highly enriched fuel, Iodine and Xenon do not significantly affect naval nuclear reactor operations while at power. However, they become a critical consideration immediately following reactor shutdown, particularly at End of Life (EOL), where the reactivity effect can increase by 50% compared to Beginning of Life (BOL).

Factors which influence this include:

- Depletion of Fuel
- Lower Differential Rod Worth.
- Larger EOL flux → Higher concentrations of Iodine → More Indirect Xe

Operators must diligently track power levels and log operating information before shutdown. This data is essential for understanding the current Xenon and Iodine concentrations in the core and predicting their impact on the next startup, which determines if the reactor is potentially "locked out". This operational challenge exists both when the ship is at sea and in port.

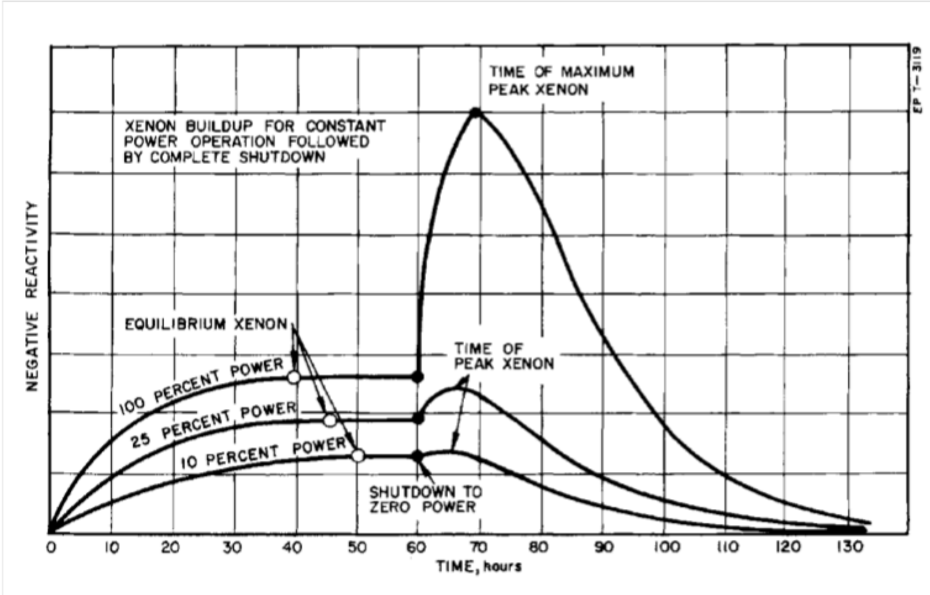

Exhibit 3-29 — Typical ^{135}Xe Buildups And Decays